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ME1100 Thermodynamics

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Thermodynamic Equilibrium & Temperature

- Thermodynamic equilibrium
  - Mechanical equilibrium
  - **Thermal equilibrium**
  - Chemical equilibrium

- When 2 bodies at different temperatures are brought into contact, after some time they attain a common temperature – they are said to be in **thermal equilibrium**

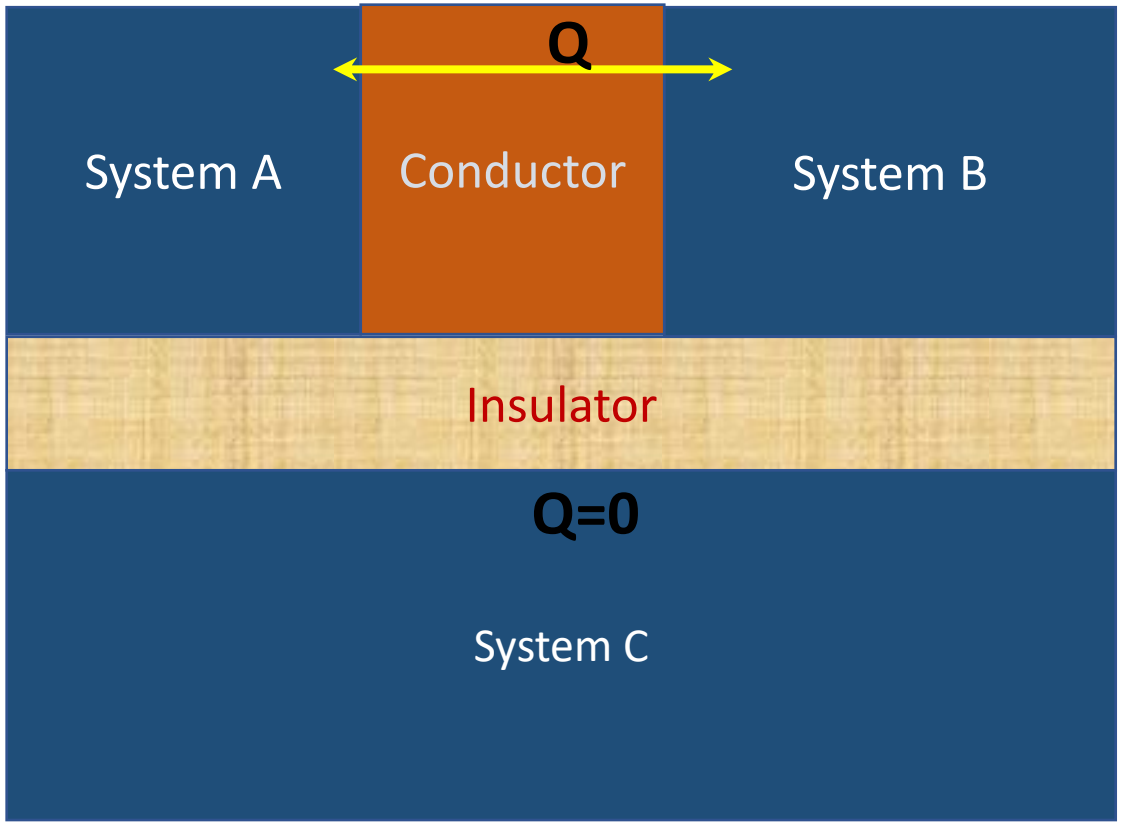
- **Temperature** may be understood as that property which determines whether two systems are in **thermal equilibrium** or not.

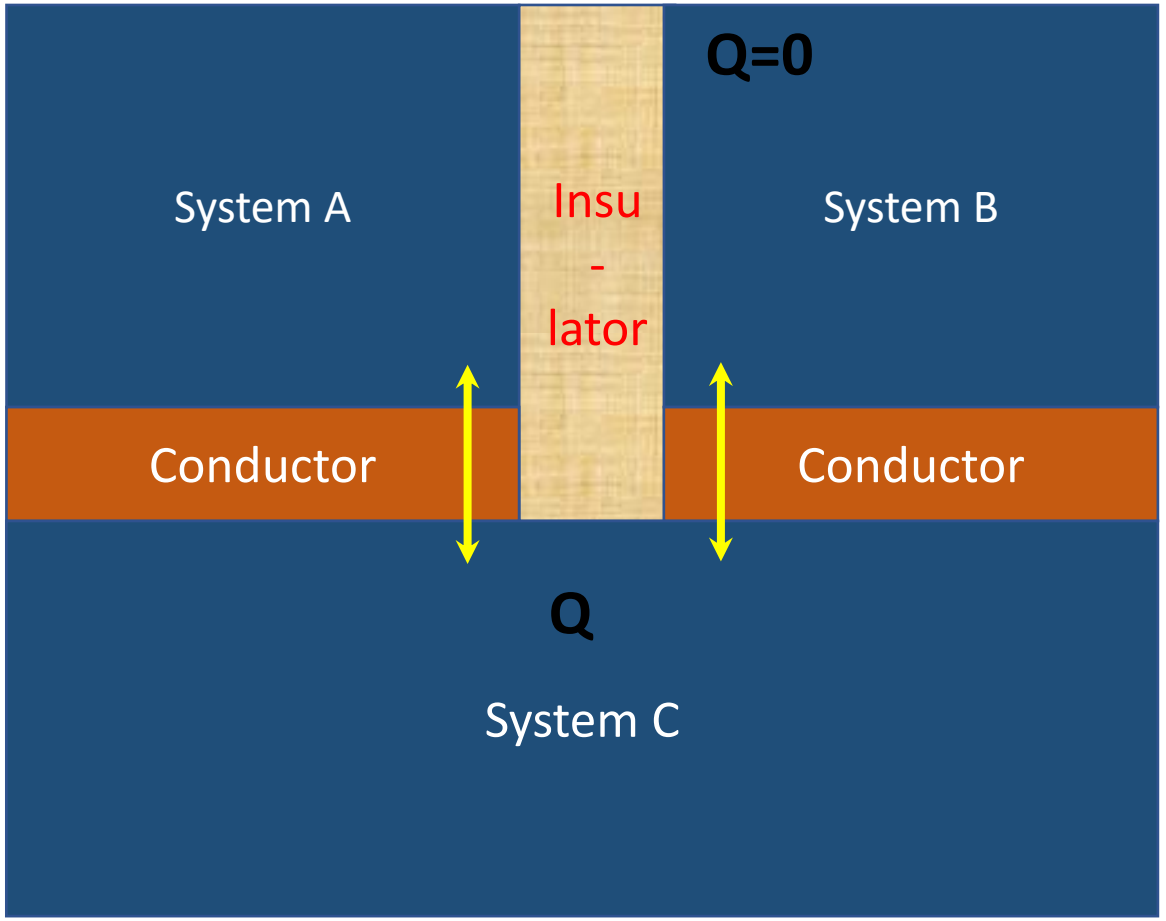
- **The property of the system by virtue of which heat flows** from one system to another— from a **system of higher temperature to a lower temperature**.

- This can be sensed and measured.

Zeroth law of Thermodynamics

Zeroth law of Thermodynamics: If **system A is in equilibrium with system B** and **system B is in equilibrium with system C**, **then systems A and C must be in equilibrium with each other**





Temperature

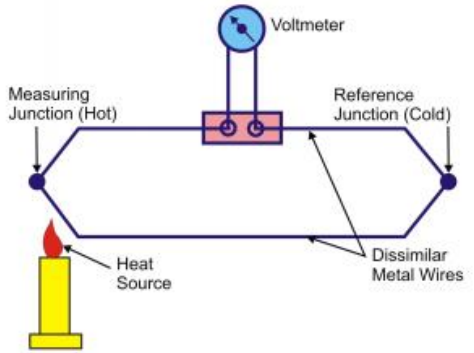
- Temperature is an intensive property and does not depend on the size of the system.
- Any property of a substance that varies with temperature can be used for measuring temperature, after proper calibration of the associated device.
- Selected characteristics are called thermometric properties and the reference body is called a thermometer.
- Thermometer is a device used to measure the temperature.
- To get a quantitative measure of temperature, a reference body is used, a certain physical characteristic of the body that changes with temperature is identified for indicating a change in temperature.
- Thermometers (thermometric properties): Constant pressure gas thermometer (V), Constant volume gas thermometer (p), Electrical resistance thermometer (R), Thermocouple (thermal-emf, e).
- Thermometers are calibrated with standard reference temperatures.

Measurement of temperature

- **The length of a copper rod changes when it is heated or cooled. The change in length is proportional to its temperature change.**
- **The change in length when the rod is brought into equilibrium with the body whose temperature is required to be measured can be used to estimate the temperature of the body**
- **The mercury-in-glass thermometer is based on the same principle, except that the temperature is calibrated in terms of the change of volume of the fluid (mercury).**
- **It is also possible to use the change of volume (or pressure) of gas with temperature to measure temperature change**
- **Ideally, we require the property to change linearly with temperature.**

Commonly used thermometers

- **Gas thermometers**
  - Constant pressure
  - Constant volume
- **Glass thermometers**
  - Mercury-in-glass
  - Alcohol-in-glass
- **Thermocouples (Seebeck effect)**
  - J (Fe-constantan CuNi), K (NiCr-NiAl), T (Cu-constantan) etc.
- **Electrical resistance thermometers**
  - Metals (Platinum, platinum-rhodium,…)
- **Liquid crystals** For liquid crystals, the color of the crystal depends on its temperature. By relating color and temperature, it is possible to use the liquid crystal as a thermometer.
- **Pyrometers**: elevated temperature measurement
  - Radiation principle, non-contact



Temperature Units

- Celsius (°C), Fahrenheit (°F), Rankine (°R), Kelvin (K)
- °F = 1.8 °C + 32
- °R = °F + 459.67
- °R = 1.8 K
- K = °C + 273.15
- **°R, K ≥ 0**

Notice that
Kelvin is K not °K

Similarly,
55 °C – 15 °C = 40 K

Temperature

Fixed point -- An easily reproducible state in an arbitrarily chosen system

Reference temperatures:

Ice point (0°C)

Steam point (100°C)

Silver melting point (961.8°C)

Gold melting point (1064°C)

Triple point of water (0.01°C). At 611.2 Pa and 0.01°C, water exits at its three phases – solid, liquid and vapor.

After 1954, triple point of water is used as a standard reference point.

If X is thermometric property at any temperature T and X_t is the value of the property measured at triple point of water, then the temperature T is evaluated as **$T = 273.16\ X / X_t$**.

Calibration of thermometers (Fixed points)

Element	TP/FP/MP	Temperature (K)
e-H2	TP	13.8033
Ne	TP	24.5561
o2	TP	54.3584
Ar	TP	83.8058
Hg	TP	234.3156
H2O	TP	273.16
Ga	MP	302.9146
Inc	FP	429.7485
Sn	FP	505.078
Zn	FP	692.677
Al	FP	933.473
Ag	FP	1234.93
Au	FP	1337.33
Cu	FP	1357.77

TP – Triple point, **MP** – Melting point at 1 atm, **FP** – Freezing point at 1 atm

Source: https://nvlpubs.nist.gov/nistpubs/Legacy/TN/nbstechnicalnote1265.pdf

Heat

- Heat is energy interaction between a system and surroundings that occurs **solely due to temperature difference between the two**
- Heat is **energy in transit**. **Heat cannot be stored !!**
- Just like work, heat interaction **between a system and surroundings** is **only defined at the system boundary**
- Heat transfer is a spontaneous process that occurs when the system and surroundings are at different temperatures.
- Both heat and work are **path functions** and are not properties.
- Heat transfer **from surroundings to a system** is considered **positive.**
- Work transfer **from system to surroundings** is considered **positive**

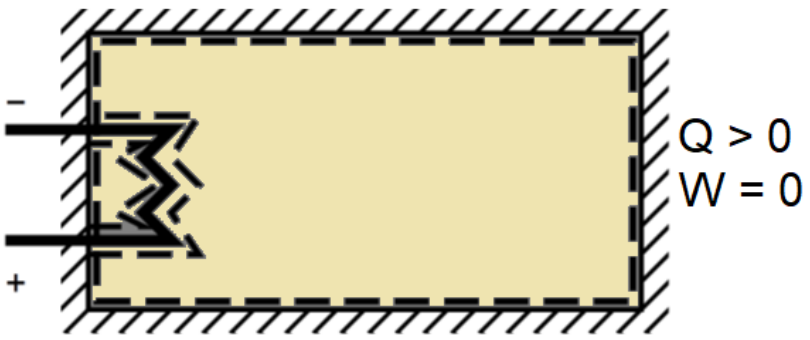
Heat

Heat transfer to a system in general raises the temperature of the system.

In phase change situations, the heat transferred to the system is utilized for heat required for phase change and temperature remains constant in those cases.

Similarly, if there is an increase in temperature that may not have resulted from heat transfer (rubbing of palms) it may result due to work transfer (movement against friction).

For an insulated system or an isolated system, there is no heat interaction. Such a system is called adiabatic.





Modes of heat transfer

- Conduction $\dot{Q} = -kA\frac{dT}{dx}$
- Convection $\dot{Q} = hA(T_1 - T_2)$
- Radiation $\dot{Q} = \sigma\epsilon A(T_1^4 - T_2^4)$
- Units of heat transfer = W (= J/s)
- $\dot{Q}$ is heat transfer rate (hence the dot on top of Q)

Thank You